

# ERD dan Ringkasan Skema — Charmbuddy

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Dokumen ini merangkum tabel utama, kolom penting, relasi (PK/FK), dan diagram ER (Mermaid) berdasarkan migration dan model di `charmbuddy-backend`.

## Ringkasan Tabel (kolom penting)

- **users**: `id` PK, `name`, `email` (unique), `role`, `avatar\_path`, `password`, `email\_verified\_at`, timestamps
- **addresses**: `id` PK, `user\_id` FK -> `users.id`, `recipient\_name`, `phone`, `address\_line`, `city`, `province`, `postal\_code`, `is\_default`, timestamps
- **categories**: `id` PK, `name`, `parent\_id` FK -> `categories.id` (nullable), timestamps
- **products**: `id` PK, `category\_id` FK -> `categories.id`, `slug` (unique), `name`, `description`, `price`, `stock`, `weight`, `image\_path`, timestamps
- **product\_images**: `id` PK, `product\_id` FK -> `products.id`, `image\_path`, timestamps
- **carts**: `id` PK, `user\_id` FK -> `users.id`, `status` (enum: active/checked\_out), timestamps
- **cart\_items**: `id` PK, `cart\_id` FK -> `carts.id`, `product\_id` FK -> `products.id`, `quantity`, unique(`cart\_id`, `product\_id`), timestamps
- **orders**: `id` PK, `user\_id` FK -> `users.id`, `cart\_id` FK -> `carts.id` (nullable), `total\_price`, `shipping\_cost`, `shipping\_address`, `courier\_service`, `status` (enum), `payment\_proof\_path`, `tracking\_number`, timestamps
- **order\_items**: `id` PK, `order\_id` FK -> `orders.id`, `product\_id` FK -> `products.id`, `quantity`, `price\_at\_checkout`, `subtotal`, timestamps
- **payments**: `id` PK, `user\_id` FK -> `users.id`, `order\_id` FK -> `orders.id`, `amount`, `status` (enum), `payment\_proof\_path`, timestamps
- **promo\_codes**: `id` PK, `code` (unique), `type` (fixed/percentage), `value`, `min\_subtotal`, `max\_discount\_amount`, `usage\_limit`, `used\_count`, `is\_active`, `starts\_at`, `ends\_at`, timestamps
- **order\_status\_histories**: `id` PK, `order\_id` FK -> `orders.id`, `status`, `note`, `changed\_by` FK -> `users.id` (nullable), `meta` (json), timestamps
- **stock\_movements**: `id` PK, `product\_id` FK -> `products.id`, `type` (in/out/adjustment), `quantity\_change`, `stock\_before`, `stock\_after`, `reason`, `source\_type`, `source\_id`, `note`, `changed\_by` FK -> `users.id` (nullable), timestamps

## Hubungan utama (ringkas)

- `users` 1---\* `carts`
- `users` 1---\* `orders`
- `users` 1---\* `addresses`
- `users` 1---\* `payments`
- `categories` 1---\* `products`
- `products` 1---\* `product\_images`
- `products` 1---\* `stock\_movements`
- `carts` 1---\* `cart\_items` \*---1 `products`
- `orders` 1---\* `order\_items` \*---1 `products`
- `orders` 1---1 `payments` (payment belongsTo order; one order may have payment record)
- `orders` 1---\* `order\_status\_histories`

## Mermaid ER diagram

```
erDiagram
    USERS {
        bigint id PK
        varchar name
        varchar email
        varchar role
```

```

}

ADDRESSES {
    bigint id PK
    bigint user_id FK
    varchar recipient_name
    varchar phone
    text address_line
}

CATEGORIES {
    bigint id PK
    varchar name
    bigint parent_id FK
}

PRODUCTS {
    bigint id PK
    bigint category_id FK
    varchar slug
    varchar name
    decimal price
    int stock
}

PRODUCT_IMAGES {
    bigint id PK
    bigint product_id FK
    varchar image_path
}

CARTS {
    bigint id PK
    bigint user_id FK
    enum status
}

CART_ITEMS {
    bigint id PK
    bigint cart_id FK
    bigint product_id FK
    int quantity
}

ORDERS {
    bigint id PK
    bigint user_id FK
    bigint cart_id FK
    decimal total_price
    varchar status
}

ORDER_ITEMS {
    bigint id PK
    bigint order_id FK
    bigint product_id FK
    int quantity
    decimal price_at_checkout
}

PAYMENTS {
    bigint id PK
    bigint user_id FK
    bigint order_id FK
    decimal amount
}

```

```

        varchar status
    }

    PROMO_CODES {
        bigint id PK
        varchar code
    }

    ORDER_STATUS_HISTORIES {
        bigint id PK
        bigint order_id FK
        varchar status
        bigint changed_by FK
    }

    STOCK_MOVEMENTS {
        bigint id PK
        bigint product_id FK
        varchar type
        int quantity_change
    }

    USERS ||--o{ ADDRESSES : "has"
    USERS ||--o{ CARTS : "owns"
    USERS ||--o{ ORDERS : "places"
    USERS ||--o{ PAYMENTS : "makes"

    CATEGORIES ||--o{ PRODUCTS : "contains"
    PRODUCTS ||--o{ PRODUCT_IMAGES : "has"
    PRODUCTS ||--o{ STOCK_MOVEMENTS : "has"

    CARTS ||--o{ CART_ITEMS : "contains"
    PRODUCTS ||--o{ CART_ITEMS : "referenced in"

    ORDERS ||--o{ ORDER_ITEMS : "contains"
    PRODUCTS ||--o{ ORDER_ITEMS : "referenced in"
    ORDERS ||--o{ ORDER_STATUS_HISTORIES : "history"
    ORDERS ||--o{ PAYMENTS : "billed"
    USERS ||--o{ ORDER_STATUS_HISTORIES : "changes"

```

Catatan: dokumen ini dibuat dari migration dan model di `charmbuddy-backend` pada repository. Jika Anda mau, saya bisa:

- mengekspor diagram sebagai gambar SVG/PNG
- membuat versi PDF/PNG di luar folder proyek
- menambahkan detail kolom (tipe lengkap, default, indeks) untuk setiap tabel

Jika setuju, saya lanjutkan mengekspor diagram ke file gambar di root workspace.